Expected Evaluation

Your evaluation must address the central question: **“Does conformance information derived from a process model improve the prediction of future deviations and compliance risks?”**

The experimental study should include:

1. A clear definition of the prediction problem and label construction: for example, defining a case as “deviant” if its final alignment cost exceeds a chosen threshold, if a particular rule is violated (e.g., an LTL constraint), or if conformance checking detects a specific type of non-compliant pattern. Motivations for the chosen definition must be documented.
2. A temporal evaluation protocol that avoids information leakage, such as splitting data into training/validation/test sets based on time or using rolling-origin evaluation. Cases must not be split across sets.
3. Training and tuning of at least one baseline model that uses no conformance-based features (only classical process mining encodings and attributes) and at least one conformance-aware model that augments these features with conformance diagnostics. Hyperparameter tuning should use a clearly described strategy (e.g., grid/random search with validation).
4. Use of appropriate evaluation metrics for the chosen target: for classification tasks, consider metrics such as ROC AUC, precision, recall, F1-score, precision–recall AUC, and calibration measures (Brier score, calibration curves); for regression-like deviation scores, use MAE, RMSE, or rank-based metrics depending on the goal. You are encouraged to analyze performance as a function of prefix length (early vs late warning).
5. Comparative analysis of baseline and conformance-aware models, including:
   – numerical comparisons (tables of metrics, confidence intervals or variability over multiple random seeds);
   – visualizations (ROC/PR curves, performance-by-prefix-length plots, feature importance analyses focusing on conformance-related variables);
   – qualitative inspection of example cases where conformance features significantly change the predicted risk.
6. Discussion of computational aspects: measure and report the runtime and resource requirements of conformance checking and feature generation, and reflect on the feasibility of deploying such a system in near real-time.
7. A critical discussion section in the report that reflects on:
   – the extent to which conformance features improved predictions in your scenario;
   – which specific conformance indicators were most useful and why;
   – potential threats to validity (quality of the process model, alignment noise, label definition);
   – limitations and ideas for future work (e.g., alternative model structures, approximate or incremental conformance, use of additional compliance languages).

The final report (approximately 10–15 pages excluding references and appendices) should connect your work to the broader literature on conformance checking and predictive process monitoring,

explaining both technical details and conceptual implications for organizations that wish to anticipate and mitigate non-compliant behavior.

References

You are encouraged to consult and cite the following (non-exhaustive) references and extend them with additional relevant work:

1. van der Aalst, W.M.P. (2016). *Process Mining: Data Science in Action* (2nd ed.). Springer.
2. Adriansyah, A., van Dongen, B.F., & van der Aalst, W.M.P. (2011). Conformance checking using cost-based fitness analysis. In *Enterprise, Business-Process and Information Systems Modeling* (BPMDS/EMMSAD).
3. Leemans, S.J.J., Fahland, D., & van der Aalst, W.M.P. (2013). Discovering block-structured process models from event logs containing infrequent behaviour. In *Business Process Management Workshops* (BPM).
4. Teinemaa, I., Dumas, M., La Rosa, M., & Maggi, F.M. (2019). Outcome-oriented predictive process monitoring: Review and benchmark. *ACM Transactions on Knowledge Discovery from Data*.
5. Verenich, I., Dumas, M., La Rosa, M., Maggi, F.M., & Teinemaa, I. (2019). Survey and cross-benchmark comparison of remaining time prediction methods in business process monitoring. *ACM Transactions on Intelligent Systems and Technology*.
6. Stevens, A., De Smedt, J., & Peeperkorn, J. (2021). Quantifying explainability in outcome-oriented predictive process monitoring. In *International Conference on Process Mining* (ICPM).

Additional literature on predictive compliance monitoring, constraint-based process modeling, and supervised process mining can be included as appropriate, provided it is properly cited in your report.